3/23/2022

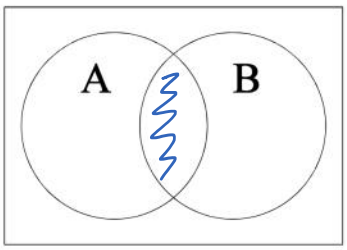
Wednesday, March 23, 2022 9:12 AM

M168_Notes_7.3

7.3 Rules of Probability

If event *A* and event *B* are mutually exclusive then they have no outcome in common.

In general, *A* and *B* have some outcomes in common so we have the union rule for probability:



Example

E is the event that the sum of the numbers shown uppermost is 7
F is the event that the red die shows a 1
G is the event that the green die shows a 6
H is the event that the sum of the numbers shown uppermost is 10

1~1	2~1	3~1	4~1	5~1	6~1
1~2	2~2	3~2	4~2	5~2	6~2
1~3	2~3	3~3	4~3	5~3	6~3
1~4	2~4	3~4	4~4	5~4	6~4
1~5	2~5	3~5	4~5	5~5	6~5
1~6	2~6	3~6	4~6	5~6	6~6

A standard deck of 52 cards has 4 suits, each with 13 cards. The suits are spades, ♠, hearts, ♥, clubs, ♣ ,and diamonds, ♦. The cards in each suit are numbered from Ace, King, Queen, Jack, ten down to 2.

Example

If a single card is drawn from a standard deck of cards, what are the probabilities of

a) a 9 or a 10? $\frac{8}{52}$

b) a black card or a 3? $\frac{28}{52}$

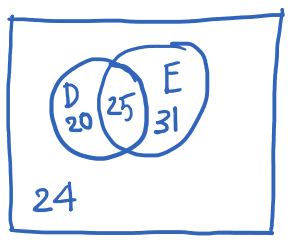
Example

A survey gave the following results: 45% of the people surveyed drank diet drinks (D) and 25% drank diet drinks and exercised ($D \cap E$) and 24% did not exercise and did not drink diet drinks ($D^c \cap E^c$). Find the probability that:

$\frac{55}{100} = 0.55$ a) a person does not drink diet drinks (D^c). $n(D^c) = 31 + 24 = 55$

b) does not exercise and drinks diet drinks ($E^c \cap D$). $n(E^c \cap D) = 20$

c) exercises and does not drink diet drinks ($E \cap D^c$). $n(E \cap D^c) = 31$



$n(D) = 45$

$n(D \cap E) = 25$

$n(D^c \cap E^c) = 24$

$x + 25 + 20 + 24 = 100$

$x = 31$